A Gauge Theory of Line Defects in an Elastic Continuum: Dislocation Rings, the Peach–Koehler Force, and Emergent Interaction Laws

(A Geometrodynamic Model of Particles as Elastic Defects)

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Abstract

We present a consistent account of a geometrodynamic model in which elementary particles are treated as topological defects (dislocation and disclination rings) of an elastic aether, and the fundamental interactions appear as different projections of a single elastic response of the medium. The model is built on the Kadić–Edelen and Lazar gauge theory of dislocations [2, 3], in which the magnetic field is identified with the dislocation density and the Peach–Koehler force plays the role of the Lorentz force. Gravity is introduced through a "Fermi film" mechanism — a deformable boundary of the action integral held orthogonal to the curved world tubes of accelerating charges (a generalization of Fermi's 1923 treatment of the 4/3 paradox [1]). We show that the inverse-square law for gravity is a *theorem* (a consequence of the Laplacian Green function), not a fit. The model is formulated in an elastic unit basis (L, T, M), without invoking electric charge as a primary entity. The limits are drawn honestly: the *structure* of the interactions (power laws, signs, hierarchy) is derived, together with a set of relations between scales, whereas the absolute values of the fundamental constants (α, G) remain input parameters. The paper is a working preprint intended as a first introduction to the subject.

Keywords: gauge theory of dislocations, aether, Fermi film, magnetic monopole, Burgers vector, Peach–Koehler force, 4/3 paradox, geometrodynamics.

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1 Introduction

1.1 Motivation: the particle as a defect of the medium

The idea that elementary particles may be not point objects but extended structures in some medium has a long history — from Kelvin's vortices to modern soliton models. The present work develops a concrete version of this programme: *a particle is a topological defect of an elastic continuum* (the "aether"), namely a closed dislocation line (ring) or its generalizations. The approach is attractive because it rests on the well-developed mathematical apparatus of the theory of crystal defects [9, 3] and its gauge formulation [2].

1.2 Why dislocations

In crystals, dislocations are line defects characterized by the Burgers vector $\mathbf{b}$, which measures the closure failure of a circuit around the defect line. M. Lazar [2] showed that the dynamic gauge theory of the translation group $T(3)$ leads to equations structurally identical to Maxwell's equations (Table 1). This provides a physically meaningful "dictionary" between defect theory and electrodynamics.

Table 1: Correspondence Maxwell $\leftrightarrow$ dislocation theory (after Lazar [2]).

Electrodynamics	Dislocation theory
Magnetic field $\mathbf{B}$	dislocation density T^a
Electric field $\mathbf{E}$	dislocation current I^a
Magnetic excitation $\mathbf{H}$	pseudomoment stress H_a
Magnetic flux $\oint \mathbf{B}\, d\mathbf{S}$	Burgers vector $\oint T^a = b$
Lorentz force	Peach–Koehler force

1.3 The key feature: the charge is magnetic, not electric

In the proposed model *electric charge is not a primary entity.* What is primary is the magnetic charge, identified with the flux of the dislocation density, i.e. with the Burgers vector. The electron is built as a precessing magnetic dipole. This is consistent with the long-standing line connecting dislocations and Dirac monopoles [10].

1.4 Structure of the paper

Section 2 introduces the elastic unit basis. Section 3 describes the particle as a precessing dipole. Section 4 presents the Fermi-film mechanism and the derivation of gravity. Section 5 collects the six interaction channels. Section 6 honestly delimits the model.

2 The elastic unit basis

We build a system of units from three primitives of the elastic medium: length L, time T and mass M. There is *no* electric charge and no ampere in the basis. The main quantities are:

- Burgers vector b (lattice displacement) — dimension $[L]$;
- shear modulus μ — dimension $[ML^{-1}T^{-2}]$ (Pa);
- density ρ — $[ML^{-3}]$;
- speed of sound/light $c = \sqrt{\mu/\rho}$ — $[LT^{-1}]$.

The magnetic charge as the flux of the dislocation density $q_m = \oint T^a\, dS$ has the dimension of length $[L]$ — this is internally consistent in the elastic theory (unlike in SI, where the magnetic charge is measured in A·m; transferring SI units to the elastic medium is unnecessary and leads to confusion).

3 The particle as a precessing dipole

3.1 Geometry

The electron is modelled as a pair of dislocation rings (of radius a) whose planes are *fixed*, while along each ring a "special point" — a localized magnetic charge $q_m = \Delta b$ (a jump of the Burgers vector) — runs with frequency ω. By "precession" we mean precisely the motion of the special point along the fixed ring, not a wobbling of the planes. The two rings are separated by a distance d and carry opposite charges $\pm q_m$ (the N and S poles), forming a magnetic dipole.

3.2 Mass from self-interaction

Mass is not introduced separately but is the elastic self-interaction energy of the defect:

$$m = \frac{\mu\, b^2 a}{c^2}, \tag{1}$$

which has the correct dimension $[M]$.

3.3 Magnetic moment and spin

The magnetic moment is a *static* dipole moment $\mu_{\mathrm{mag}} = q_m\, d$ (not a current-loop moment: the ring current is magnetic, and the naive current-loop formula does not apply here). Spin is tied to topology: a pair of ordinary rings has configuration period 2π (a boson), whereas a pair of Möbius rings (ring + anti-ring) has period 4π, corresponding to a fermion of spin 1/2. In the Möbius pair the two rings carry *opposite* directions of the Burgers vector ($+\Delta b$ and $-\Delta b$, the N and S poles) — hence a static magnetic dipole. Crucially, the Burgers twist $b(\varphi) = \cos(\varphi/2)$ is a Frank rotation by π per circuit, i.e. a *disclination* ($SO(3)$) degree of freedom: the Möbius sector of the model already lies in $SO(3)$, and the magnetic moment $\mu = \nu = n/2$ follows as the flux of the spin connection (Osipov [6]). Hydrodynamically the same object is a vortex: the vorticity corresponds to the Frank density, and the precession ($\omega > 0$) required for stability is the circulation of that vortex.

3.4 Stability and annihilation

Opposite charges $\pm q_m$ attract and, without motion, would collapse. Precession provides a centrifugal barrier (conservation of angular momentum $J = ma^2\omega$) that stabilizes the pair at a finite distance:

$$E(d) = -\frac{A}{d} + \frac{J^2}{2md^2}, \qquad d_{\mathrm{eq}} = \frac{J^2}{mA}. \tag{2}$$

As $\omega \to 0$ the barrier vanishes — the pair annihilates (Frenkel recombination, the total Burgers vector going to zero).

4 The Fermi film and the derivation of gravity

4.1 Historical context: the $4/3$ paradox

E. Fermi in 1923 [1] showed that the well-known discrepancy between the electrodynamic ($\frac{4}{3}U/c^2$) and relativistic (U/c^2) masses of a charged body is removed if the boundary of the action integral is taken *strictly orthogonal* to the curved world tube of the accelerating charge. We call this deformable boundary the *Fermi film*. The film is not a term of the Lagrangian but the boundary of the integration domain of the action. It is therefore not a dynamical field, is not retarded, and adjusts instantaneously to the curvature of the world tubes. Only the field itself, obtained by minimizing the action, is retarded and radiates. The Fermi film should be distinguished from the Gibbons–Hawking–York boundary term of general relativity [8]. Our film differs from it in that it does not enter the Lagrangian: the Gibbons–Hawking–York term is a term of the action and obeys the field dynamics, whereas the Fermi film is merely the boundary of the integration domain. Therefore it is not retarded and adjusts instantaneously to the curvature of the world tubes.

4.2 The source of deformation

An accelerating charge curves its world tube; the film, remaining orthogonal, is deformed. The density of the deformation source is proportional to the *square* of the proper acceleration $W = \omega^2 a$:

$$\sigma = \mu\, b^2\, \frac{\langle W^2 \rangle}{c^2} > 0. \tag{3}$$

The sign is always positive (a square), so gravity is always attractive and does not depend on the sign of the charge (universality).

4.3 The inverse-square law as a theorem

Theorem 1 ($1/R^2$ law)**.** *If the film deformation Δ obeys the equilibrium equation of an elastic membrane with a point source,*

$$\gamma\, \nabla^2 \Delta(\mathbf{r}) = -\sigma\, \delta^3(\mathbf{r} - \mathbf{r}_0), \tag{4}$$

then the solution is the Laplacian Green function, $\Delta(R) = \sigma/(4\pi\gamma R)$, whence the interaction potential $U(R) \propto 1/R$ and the force $F(R) \propto 1/R^2$.

The exponent follows from the three-dimensionality of space, not from a fit.

4.4 Stiffness parameters: two scales

What μ_{aeth} is. The quantity μ_{aeth} is the *shear modulus of the vacuum*, the stiffness of the aether with respect to shear deformation, of pressure dimension [Pa]. It is through

this elastic modulus that, in the (L, T, M) basis (without the ampere), the main quantities of the model are defined:

$$c = \sqrt{\mu_{\text{aeth}}/\rho} \qquad \text{(transverse-wave speed)}, \tag{5}$$

$$m = \mu_{\text{aeth}}\, b^2 a/c^2 \qquad \text{(particle mass)}, \tag{6}$$

$$F = \mu_{\text{aeth}}\, b^2/R \cdot (\text{geometry}) \qquad \text{(Coulomb force)}. \tag{7}$$

Physically μ_{aeth} governs shear deformations (dislocations), from which charge and mass are built; hence it sets the physics at the particle scale (of order the Compton length λ_C). Its numerical value is obtained by *calibration*: substituting the electron mass m_e and its classical radius r_e into the self-energy of the dislocation ring and solving for μ_{aeth}. This is *not* a derivation from first principles but the fixing of a single number by the electron, which we state explicitly. A later analysis of the mass (inertia concentrated in the special points, their energy set by Lazar's modulus c_1) showed that this value is only an order-of-magnitude estimate; see the extended version.

A check against the model shows that *one* modulus cannot account for both mass/Coulomb and gravity: the required values differ by $\sim$84 orders of magnitude. Hence two scales are needed:

$$\mu_{\text{aeth}} \approx 2.7 \times 10^{28}\ \text{Pa} \qquad \text{(EM/mass, Compton scale)}, \tag{8}$$

$$\gamma_P = \frac{c^7}{8\pi\hbar G^2} \approx 1.8 \times 10^{112}\ \text{Pa} \qquad \text{(gravity, Planck scale)}. \tag{9}$$

The gravitational constant G enters γ_P: it is an *input* parameter of the aether stiffness, not derived from more fundamental quantities.

What γ_P is, and the nature of the hierarchy. The Planck modulus is the Planck *energy density* divided by 8π:

$$\gamma_P = \frac{c^7}{8\pi\hbar G^2} = \frac{1}{8\pi}\,\frac{m_P c^2}{\ell_P^3}, \qquad \ell_P = \sqrt{\hbar G/c^3}, \quad m_P = \sqrt{\hbar c/G} \tag{10}$$

(numerical agreement $< 1\%$). This is not an arbitrary combination but the stiffness of the vacuum at the Planck scale; the factor 8π is the same as in the Einstein equations. Likewise $\mu_{\text{aeth}} = m_e c^2/(r_e^2 \lambda_C)$ is the energy density at the electron scale. The ratio of the two moduli

$$\frac{\gamma_P}{\mu_{\text{aeth}}} \sim \frac{m_P}{m_e}\ (\text{with a volume factor}) \sim 10^{84} \tag{11}$$

reduces to the ratio of the Planck and Compton scales, i.e. to the smallness of m_e/m_P. Thus the hierarchy of the two stiffnesses is *equivalent* to the standard *hierarchy problem* ($m_e \ll m_P$): the model does not solve it, but neither does it introduce a new arbitrary parameter — it inherits the well-known smallness of m_e/m_P.

5 The six interaction channels

All interactions are projections of a single elastic response. Their decay laws and sources are collected in Table 2; the power laws are structural consequences (Green function, dipole–dipole interaction, mediator mass, confinement), not the result of a fit.

Table 2: The six interaction channels of the model.

Channel	Law	Source	Coupling
Coulomb	$1/R^2$	product of Burgers charges	$\alpha = 7.30 \times 10^{-3}$
Magnetic	$1/R^4$	dipole–dipole $q_m d$	$\sim \alpha/4$
Gravity	$1/R^2$	$\langle W^2 \rangle$ of the Fermi film	$\alpha_G = 1.75 \times 10^{-45}$
Exchange	$1/R^3$	precession phase $\cos 2\Delta\phi$	$\sim \alpha$
Weak	e^{-R/R_W}	twist chirality	$\sim 1/30$
Strong	$\propto R$	fractional $b/3$ + ribbon	~ 1

The whole system closes on two dimensionless numbers:

$$\alpha = \frac{ke^2}{\hbar c} = \frac{1}{137.04}, \qquad \alpha_G = \left(\frac{m_e}{m_{\text{Planck}}}\right)^2 = 1.75 \times 10^{-45}, \tag{12}$$

whence the weakness of gravity relative to the Coulomb force: $\alpha_G/\alpha \approx 2.4 \times 10^{-43}$.

6 Limits of the model: what is proved, assumed, open

In the spirit of scientific honesty we explicitly separate the levels of certainty.

Proved (structurally). The decay laws of all channels; the attractiveness and universality of gravity; the dimensional closure of mass; the consistency of the magnetic charge as a length-dimension quantity in the elastic basis; the annihilation mechanism and the protection by precession; the Frank vector $\Omega = \pi$ of the $\cos(\varphi/2)$ twist (a wedge half-disclination, $\nu = 1/2$, spin 1/2).

Assumed (input parameters). The constants α and α_G; the gravitational constant G (embedded in the stiffness γ_P); the existence of two stiffness scales and their ratio.

Open. The origin of the numerical values of α and α_G; a rigorous derivation of the gyromagnetic ratio $g = 2$ (in the classical ring picture one gets $g = 1$; $g = 2$ requires a spinor level of description in the spirit of [6]); the dimensional factor in the gravity source, which requires a dynamical (time-dependent) film stiffness; the numerical values of the weak- and strong-sector couplings. The tensor ($J = 2$) channel of the trace-free quadrupole Q_{ij} exists structurally (spin-2) but gives only a short-range $1/R^5$ correction; the long-range $1/R^2$ gravity is intrinsically scalar (the trace $\langle W^2 \rangle$).

7 Conclusion

We have proposed an internally consistent model in which particles are elastic defects of the aether, and the interactions are projections of a single medium response through the Peach–Koehler force and the deformation of the Fermi film. The model explains the *structure* (laws, signs, hierarchy of interactions) without claiming to derive the fundamental constants from first principles. Such a formulation makes the theory falsifiable: for magneto-gravitational devices, for example, it predicts a negligible ($\sim 10^{-13}$) relative

correction to weight, incompatible with claims of macroscopic antigravity. Further work includes a rigorous spinor derivation of $g = 2$, a refinement of the dynamical film stiffness, and this English version for archival publication.

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